

Course name: Data Communication LAB [K], Course code: COE 3201

Semester: Spring 2025-2026, Name of the Faculty: Dr. Muhammad Morshed Alam

Experiment no: 02

Experiment name: Study of signal frequency, spectrum, bandwidth, bit rate, quantization using MATLAB

Group no: 05

SL	Student name	ID	Marks
1	Maisha Mahjabin	23-54978-3	
2	Israt Jahan Surovy	23-54972-3	
3	Chowdhury Zarin Tasnim Oishy	23-54720-3	
4	Tanjimul Hossain Mojumder	23-52826-2	
5	Showmikul Islam Lemo	23-52830-2	

Marking Rubrics (to be filled by Faculty):

Category	Proficient [5]	Good [4]	Acceptable [3-2]	Unacceptable [1]	Secure d Marks
Objective/ related theory	Explains all necessary and relevant theoretical background information, measures, and variables.	Explains the important parts of theoretical background information, may be partially irrelevant or need more details.	Explains some theoretical background information, but some information may be missing, irrelevant, or inaccurate.	A lot of information is missing, out of context and/or inaccurate.	
Procedure/code	Working procedure or code is clearly presented and is supported	Working procedure or code is given for the experime	Working procedure or code is missing some steps and/or contains	Working procedure or code is absent or missing major steps	

	by proper comments. • Methods are clearly written, including all steps in sufficient detail for the experiment to be repeated.	nt to be sufficient. • Methods are correct but the steps may be lacking in detail, making the experiment hard to be repeated.	some mistakes. • Simulation output has many problems.	and/or contains mistakes. • Simulation output completely wrong.	
Experiment results/table/simulation output figures	• Clear, accurate diagrams and graphs are labeled neatly and accurately with excellent detail. • Simulated result meets all criteria; outcomes are described clearly and accurately.	• Diagrams and plots are included and are correctly labeled in brief, but there may be some lack of clarity. • Most criteria are met, but there may be some lack of clarity and/or incorrect information.	• Diagrams and plots are included and are labeled, minor mistakes may be present. • Results do not match exactly with the theoretical values and/or analysis is unclear.	• Needed plots/diagrams are missing or are missing important labels. • Experimental results are missing or completely incorrect.	
Data Interpretation/analysis/question answer/conclusion	• Interpretation and analysis of related outcomes (consequences and	• Analysis is logically tied to information; some related	• Analysis is inconsistently tied to some of the information discussed; related	• Only the data was reported, there is no analysis. • Report questions	

	implications) are logical. Any report questions are properly answered with detailed justification or calculations.	outcomes are not clear. Report questions are answered correctly but may be lacking detail or contain minor logical error.	outcomes are oversimplified. Report questions are answered but contain wrong information or major logical error.	are missing or are completely wrong.	
Comments:				Total Marks (Out of 20):	

Objectives:

This experiment is designed to understand the use of MATLAB for solving communication engineering problems and to develop understanding of MATLAB environment, commands and syntax.

Theory:

Frequency: Frequency is the number of times a periodic signal repeats per second, measured in Hertz (Hz). It determines how fast a signal oscillates. For example, a 500 Hz signal completes 500 cycles in one second.

Spectrum: The spectrum of a signal shows the signal's composition in terms of different frequency components. It is obtained using Fourier Transform, and it tells us how much of each frequency is present in the signal.

Bandwidth: Bandwidth is the range of frequencies over which a signal has significant energy. It indicates the width of the spectrum that the signal occupies. A wider bandwidth usually means the signal changes faster over time.

Quantization: It is the process of mapping a continuous range of signal amplitudes to a finite set of discrete levels. It is used in digital signal processing to represent analog signals with a limited number of levels, introducing a small approximation error.

Code:

a) ID: 23-54972-3

```
clc;
clear;
A1 = 24;      % GD
A2 = 27;      % AF
f1 = 500;     % C*100
f2 = 700;     % F*100
```

b)

```

T1 = 1/f1;
T2 = 1/f2;
fs = 20000;    % Sampling frequency
cycle = 2;     % Number of cycles to plot
t = -T1 : 1/fs : cycle*T2;

x1 = A1*cos(2*pi*f1*t);
x2 = A2*cos(2*pi*f2*t);
x3 = x1 + x2;

subplot(3,1,1)
plot(t, x3, 'LineWidth', 2)
xlabel('Time (sec)')
ylabel('x3 (volts)')
title('Task (b): Time domain representation of x3')
grid on

```

c)

```

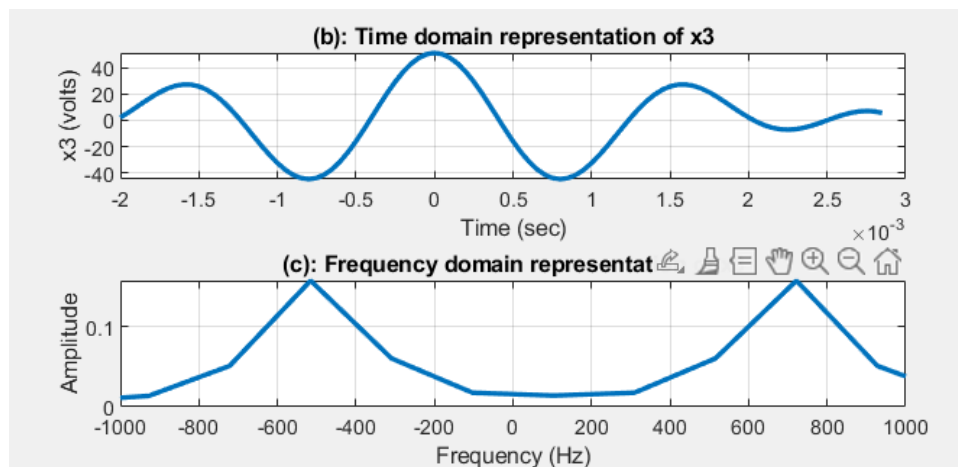
N3 = length(x3);
f_axis = linspace(-1,1,N3)*(fs/2);
X3 = fftshift(fft(x3)/(fs/2));

subplot(3,1,2)
plot(f_axis, abs(X3), 'LineWidth', 2)
xlim([-1000 1000])
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Task (c): Frequency domain representation of x3')
grid on

% Bandwidth
bw_x3 = obw(x3, fs);
fprintf('The Bandwidth of x3 is: %.2f Hz\n', bw_x3);

```

Result:



```
The Bandwidth of x3 is: 914.26 Hz
fx >>
```

Discussion:

In this experiment, a composite signal

$$x_3 = 24 \cos(2\pi 500t) + 27 \cos(2\pi 700t)$$

was generated by combining two cosine signals with frequencies 500 Hz and 700 Hz. Since both signals are periodic, their sum is also periodic. The time-domain plot displayed approximately two cycles starting from a negative time value to include $t=0$. A sampling frequency of 20,000 Hz was used to ensure sufficient samples per period and avoid aliasing. In the frequency domain, two clear peaks were observed at 500 Hz and 700 Hz, confirming the presence of the original frequency components. The theoretical bandwidth of the signal is:

$$700 - 500 = 200 \text{ Hz}$$

However, MATLAB's `obw()` function produced a bandwidth of approximately 914 Hz. This occurred because a short signal duration was used for FFT analysis. A short observation time reduces frequency resolution, causing spectral leakage and spreading of energy across nearby frequencies. Since `obw()` measures occupied bandwidth (99% power), the measured bandwidth became larger than the theoretical value.

Question and Answer:

$$x_1 = 50 \sin(10\pi t)$$

$$x_2 = 25 \sin(20\pi t)$$

$$x = x_1 + x_2$$

$$nx = x + \text{noise}$$

$$\text{noise} = 5 * \text{randn}(\text{size}(x))$$

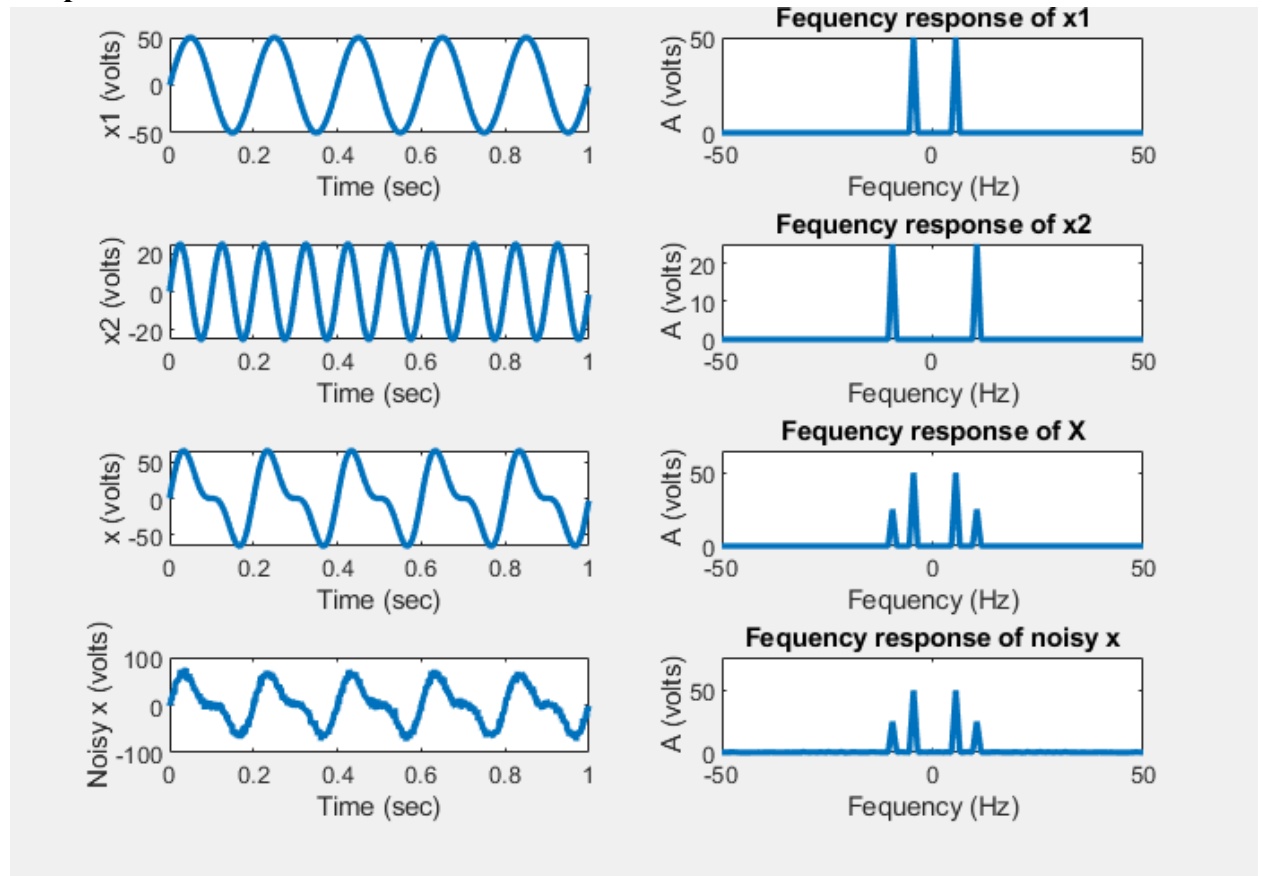
1. Subplot (4,2) to plot signal in time domain and frequency domain
2. Calculate the bandwidth of the signal x using “obw” MATLAB built-in function

1) Code:

```
3. clc; clear;
4. fs=1000;
5. t=0:1/fs:1-1/fs; % time domain
6. x1=50.*sin(2*pi*5*t);
7. x2=25.*sin(2*pi*10*t);
8. x=x1+x2; % composite signal
9. n=5.*randn(size(x));
10.    nx=x+n; % noisy x (composite signal)
11.
12.    bw=obw(x,fs) % calculating signal bandwidth
13.
14.    %frequency domain
15.    f=linspace(-1,1,fs).*(fs/2);
16.
17.    fx1=fft(x1); % fast fourier transform to get the frequency
    response
18.    fx1=fftshift(fx1)/(fs/2); % frequency domain of x1
19.    fx2=fft(x2);
20.    fx2=fftshift(fx2)/(fs/2);
21.    fx=fft(x);
22.    fx=fftshift(fx)/(fs/2);
23.    fnx=fft(nx);
24.    fnx=fftshift(fnx)/(fs/2);
25.
26.    subplot (4,2,1)
27.    plot(t,x1,LineWidth=2)
28.    xlabel('Time (sec)')
29.    ylabel ('x1 (volts)')
30.
31.    subplot (4,2,2)
32.    plot(f,abs(fx1),LineWidth=2)
33.    xlabel('Frequency (Hz)')
34.    ylabel ('A (volts)')
35.    title('Frequency response of x1')
36.    axis([-50 50 0 max(x1)])
37.
38.    subplot (4,2,3)
39.    plot(t,x2,LineWidth=2)
40.    xlabel('Time (sec)')
41.    ylabel ('x2 (volts)')
42.
43.    subplot (4,2,4)
44.    plot(f,abs(fx2),LineWidth=2)
45.    xlabel('Frequency (Hz)')
```

```
46.     ylabel ('A (volts)')
47.     title('Fequency response of x2')
48.     axis([-50 50 0 max(x2)])
49.
50.     subplot (4,2,5)
51.     plot(t,x,LineWidth=2)
52.     xlabel('Time (sec)')
53.     ylabel ('x (volts)')
54.
55.     subplot (4,2,6)
56.     plot(f,abs(fx),LineWidth=2)
57.     xlabel('Fequency (Hz)')
58.     ylabel ('A (volts)')
59.     title('Fequency response of X')
60.     axis([-50 50 0 max(x)])
61.
62.
63.     subplot (4,2,7)
64.     plot(t,nx,LineWidth=2)
65.     xlabel('Time (sec)')
66.     ylabel ('Noisy x (volts)')
67.
68.     subplot (4,2,8)
69.     plot(f,abs(fnx),LineWidth=2)
70.     xlabel('Fequency (Hz)')
71.     ylabel ('A (volts)')
72.     title('Fequency response of noisy x')
73.     axis([-50 50 0 max(nx)])
```


Output:



2)

bw =

5.9688

f_x >>

Conclusion:

This experiment successfully demonstrated both time-domain and frequency-domain analysis of a composite signal. Overall, the experiment reinforces fundamental concepts of frequency analysis, spectral behavior, bandwidth estimation, spectral leakage, and quantization in signal processing. These principles are essential for understanding communication systems and digital signal processing applications.

References:

[1] MathWorks, *MATLAB Documentation*, Natick, MA, USA: MathWorks. Online. Available: <https://www.mathworks.com/help/>

[2] Stormy Attaway, *MATLAB: A Practical Introduction to Programming and Problem Solving*, 6th ed. Boston, MA, USA: Butterworth-Heinemann, 2019.